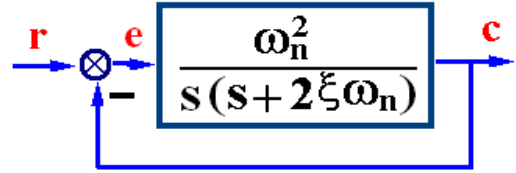


§ 3.3.3 欠阻尼二阶系统动态性能指标计算

$$G(s) = \frac{\omega_n^2}{s(s + 2\xi\omega_n)}$$



$$\Phi(s) = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2} \quad (0 \leq \xi < 1)$$

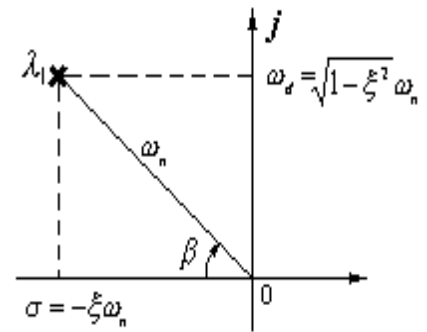
1. 欠阻尼二阶系统极点的两种表示方法

(1) 直角坐标表示

$$\lambda_{1,2} = \sigma \pm j\omega_d = -\xi\omega_n \pm j\sqrt{1-\xi^2}\omega_n$$

(2) “极”坐标表示

$$\begin{cases} |\lambda| = \omega_n \\ \angle \lambda = \beta \end{cases} \quad \begin{cases} \cos \beta = \xi \\ \sin \beta = \sqrt{1-\xi^2} \end{cases}$$



欠阻尼二阶系统极点表示

2. 欠阻尼二阶系统的单位阶跃响应

$$\begin{aligned} C(s) &= \Phi(s)R(s) = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2} \cdot \frac{1}{s} \\ &= \frac{[s^2 + 2\xi\omega_n s + \omega_n^2] - s(s + 2\xi\omega_n)}{s(s^2 + 2\xi\omega_n s + \omega_n^2)} \\ &= \frac{1}{s} - \frac{s + 2\xi\omega_n}{(s + \xi\omega_n)^2 + (1 - \xi^2)\omega_n^2} \end{aligned}$$

$$= \frac{1}{s} - \frac{s + \xi \omega_n}{(s + \xi \omega_n)^2 + (1 - \xi^2) \omega_n^2} - \frac{\xi}{\sqrt{1 - \xi^2}} \cdot \frac{\sqrt{1 - \xi^2} \omega_n}{(s + \xi \omega_n)^2 + (1 - \xi^2) \omega_n^2}$$

利用复位移定理 $L[f(t) \cdot e^{-at}] = F(s + a)$

系统单位阶跃响应为

$$h(t) = 1 - e^{-\xi \omega_n t} \cos \sqrt{1 - \xi^2} \omega_n t - \frac{\xi}{\sqrt{1 - \xi^2}} e^{-\xi \omega_n t} \sin \sqrt{1 - \xi^2} \omega_n t$$

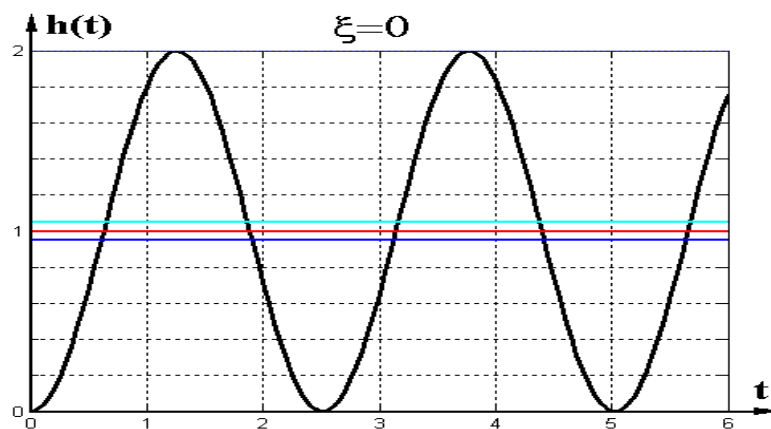
$$= 1 - \frac{e^{-\xi \omega_n t}}{\sqrt{1 - \xi^2}} \left[\sqrt{1 - \xi^2} \cos \sqrt{1 - \xi^2} \omega_n t + \xi \sin \sqrt{1 - \xi^2} \omega_n t \right]$$

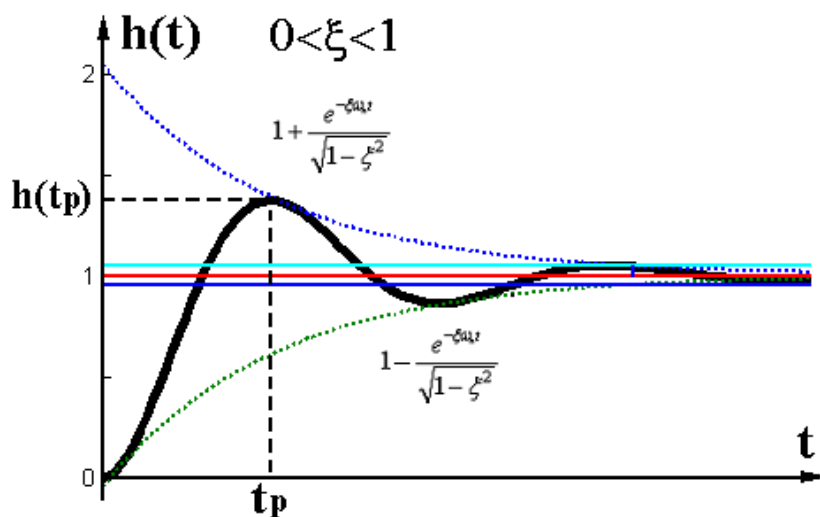
$$= 1 - \frac{e^{-\xi \omega_n t}}{\sqrt{1 - \xi^2}} \left[\sin \beta \cdot \cos \sqrt{1 - \xi^2} \omega_n t + \cos \beta \cdot \sin \sqrt{1 - \xi^2} \omega_n t \right]$$

$$h(t) = 1 - \frac{e^{-\xi \omega_n t}}{\sqrt{1 - \xi^2}} \sin(\sqrt{1 - \xi^2} \omega_n t + \beta) \quad (0 \leq \xi < 1)$$

$$h(t) = 1 - \sin(\omega_n t + 90^\circ) = 1 - \cos \omega_n t$$

$$(\xi = 0, \beta = 90^\circ)$$





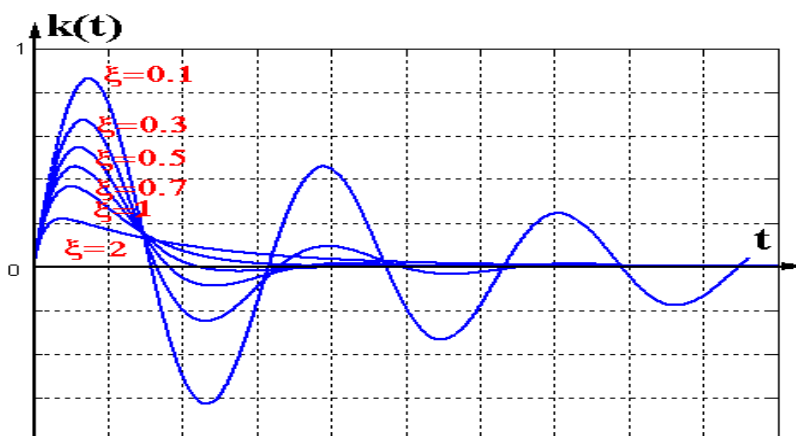
$$h(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \sin(\sqrt{1-\xi^2}\omega_n t + \beta) \quad (0 \leq \xi < 1)$$

系统单位脉冲响应为

$$k(t) = h'(t) = L^{-1}[\Phi(s)] = L^{-1}\left[\frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}\right]$$

$$= L^{-1}\left[\frac{\omega_n}{\sqrt{1-\xi^2}} \cdot \frac{\sqrt{1-\xi^2}\omega_n}{(s + \xi\omega_n)^2 + (1-\xi^2)\omega_n^2}\right]$$

$$= \frac{\omega_n}{\sqrt{1-\xi^2}} e^{-\xi\omega_n t} \sin \sqrt{1-\xi^2}\omega_n t$$



欠阻尼二阶系统单位脉冲响应

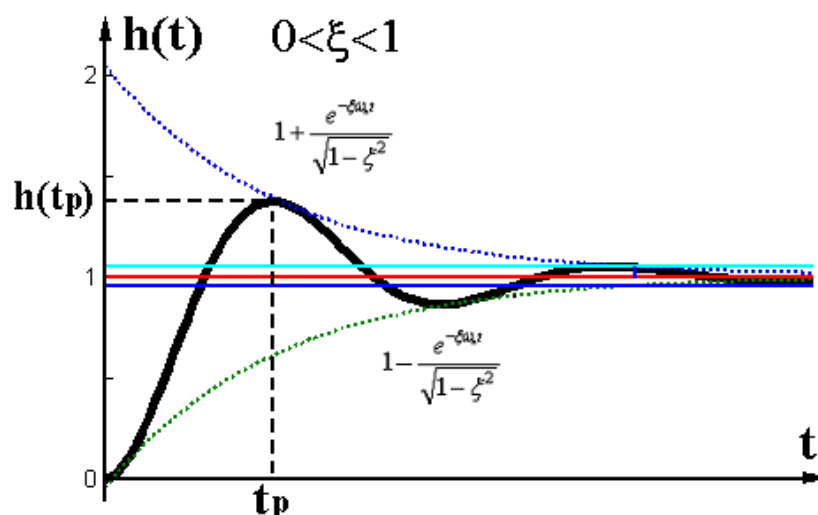
3. 欠阻尼二阶系统动态性能指标计算

(1) 峰值时间 t_p

$$h'(t) = k(t) = \frac{\omega_n}{\sqrt{1-\xi^2}} e^{-\xi\omega_n t} \sin \sqrt{1-\xi^2} \omega_n t = 0$$

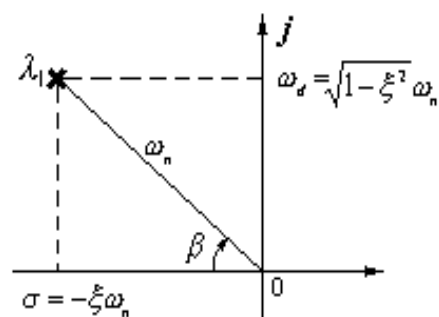
$$\sin \sqrt{1-\xi^2} \omega_n t = 0$$

$$\sqrt{1-\xi^2} \omega_n t = 0, \pi, 2\pi, 3\pi, \dots$$



由峰值时间定义

$$t_p = \frac{\pi}{\sqrt{1-\xi^2} \omega_n}$$



欠阻尼二阶系统极点表示

(2) 超调量 $\sigma\%$

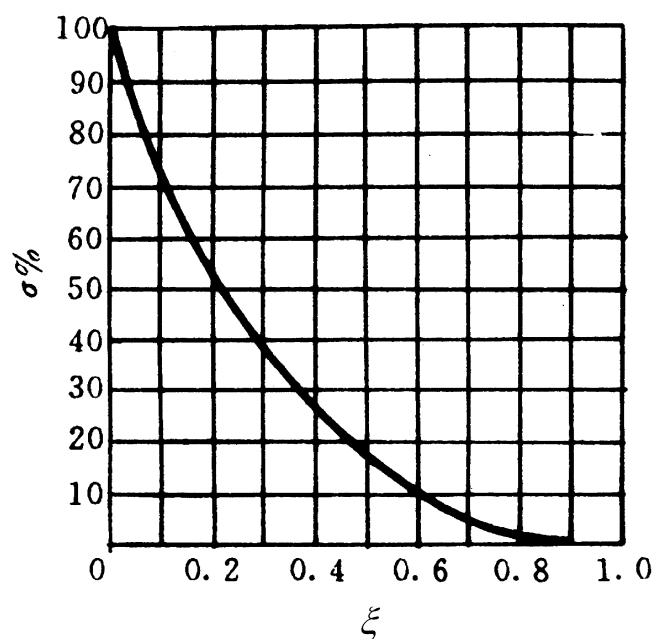
$$h(t_p) = 1 - \frac{e^{-\xi\omega_n t_p}}{\sqrt{1-\xi^2}} \sin(\sqrt{1-\xi^2}\omega_n t_p + \beta)$$

$$= 1 - \frac{e^{-\xi\pi/\sqrt{1-\xi^2}}}{\sqrt{1-\xi^2}} \sin(\pi + \beta)$$

$$= 1 + e^{-\xi\pi/\sqrt{1-\xi^2}}$$

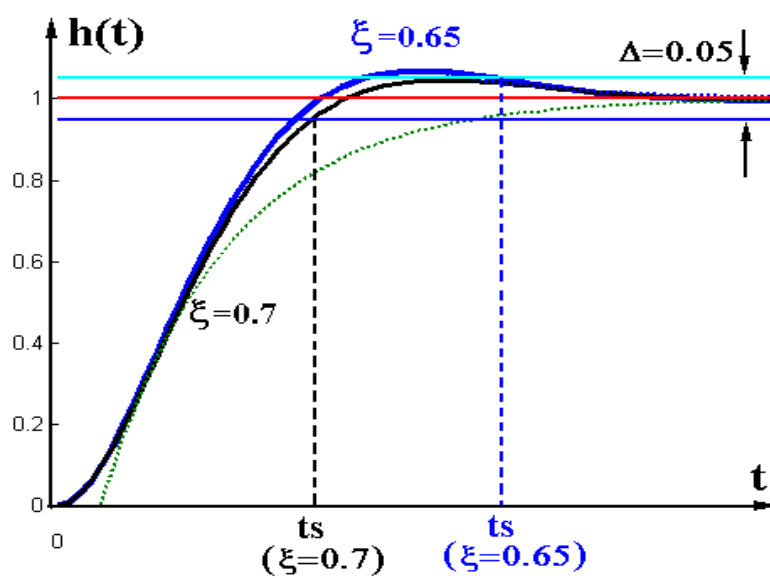
$$\sigma\% = \frac{h(t_p) - h(\infty)}{h(\infty)} \times 100\% = e^{-\xi\pi/\sqrt{1-\xi^2}} \times 100\%$$

$$\sigma\% = e^{-\xi\pi/\sqrt{1-\xi^2}} \times 100\%$$

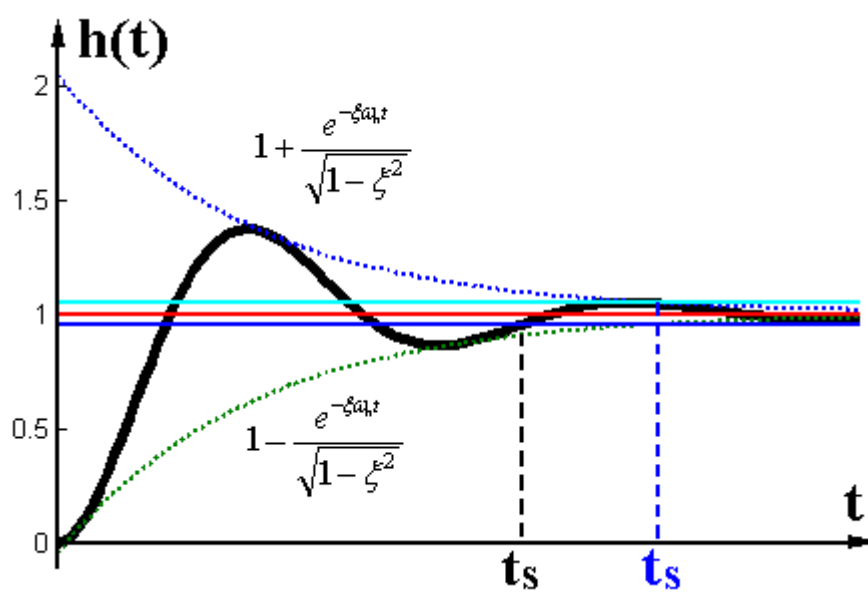


超调量 $\sigma\%$ 与阻尼比 ξ 之间的关系

(3) 调节时间 t_s



t_s 对 ξ 的不连续性



调节时间的实际计算方法

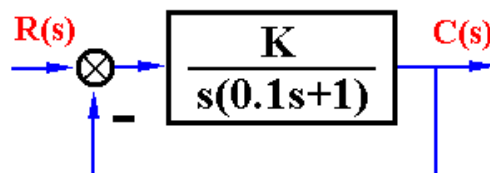
$$\left| 1 + \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} - 1 \right| = \frac{e^{-\xi\omega_n t_s}}{\sqrt{1-\xi^2}} = 0.05$$

$$t_s = -\frac{\ln 0.05 + \frac{1}{2} \ln(1 - \xi^2)}{\xi \omega_n} \approx \frac{3.5}{\xi \omega_n} \quad (0.3 < \xi < 0.8)$$

$$t_s = \frac{3.5}{\xi \omega_n} \quad (\Delta\% = 5\%)$$

$$\left\{ \begin{array}{l} t_p = \frac{\pi}{\sqrt{1 - \xi^2} \omega_n} \\ \sigma\% = e^{-\xi\pi/\sqrt{1 - \xi^2}} \times 100\% \\ t_s = \frac{3.5}{\xi \omega_n} \quad (\Delta\% = 5\%) \end{array} \right.$$

例1 控制系统结构图如图所示



- (1) 开环增益 $K = 10$ 时, 求系统的动态性能指标;
- (2) 确定使系统阻尼比 $\xi = 0.707$ 的 K 值。

解 (1) $G(s) = \frac{K}{s(0.1s + 1)}$

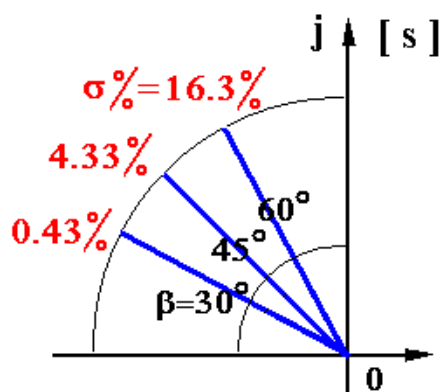
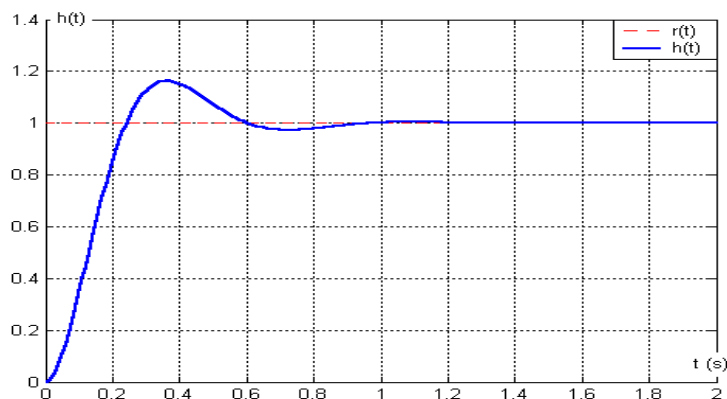
$K = 10$ 时 $\Phi(s) = \frac{G(s)}{1 + G(s)} = \frac{100}{s^2 + 10s + 100} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$

$$\begin{cases} \omega_n = \sqrt{100} = 10 \\ \xi = \frac{10}{2 \times 10} = 0.5 \quad (\beta = 60^\circ) \end{cases}$$

$$t_p = \frac{\pi}{\sqrt{1-\xi^2}\omega_n} = \frac{\pi}{\sqrt{1-0.5^2} \times 10} = 0.363$$

$$\sigma \% = e^{-\xi\pi/\sqrt{1-\xi^2}} = e^{-0.5\pi/\sqrt{1-0.5^2}} = 16.3\%$$

$$t_s = \frac{3.5}{\xi\omega_n} = \frac{3.5}{0.5 \times 10} = 0.7$$



(2) $\Phi(s) = \frac{10K}{s^2 + 10s + 10K}$

$$\begin{cases} \omega_n = \sqrt{10K} \\ \xi = \frac{10}{2\sqrt{10K}} \end{cases}$$

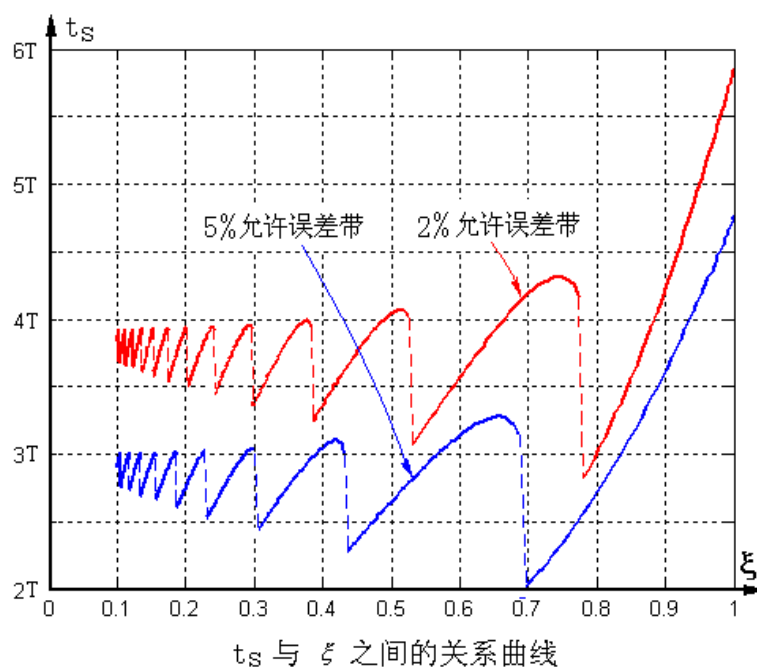
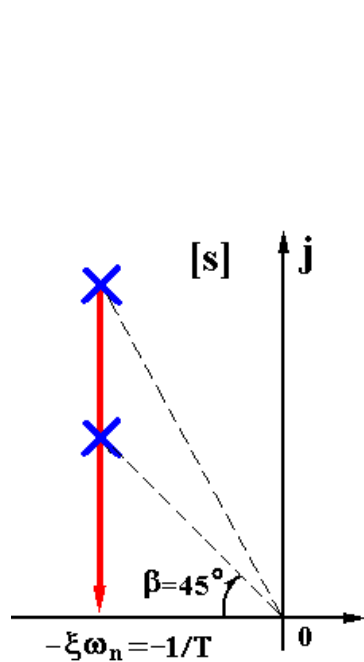
令 $\xi = 0.707$

得

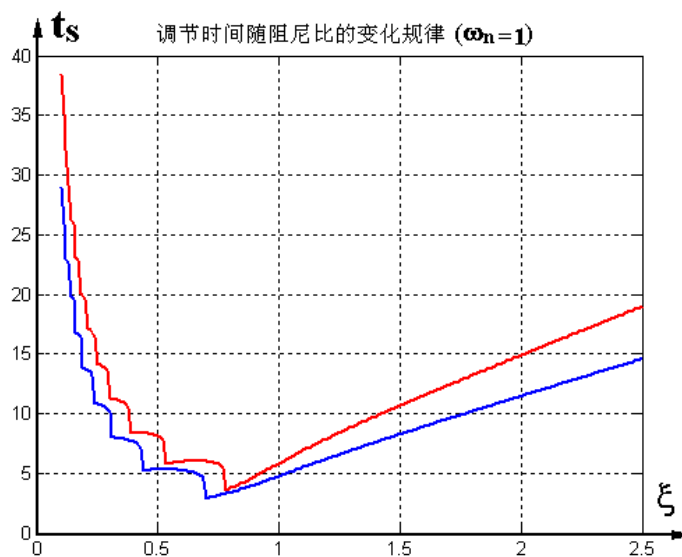
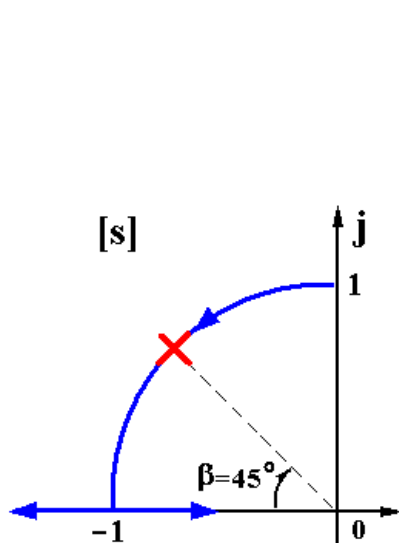
$$K = \frac{100 \times 2}{4 \times 10} = 5$$

4. “最佳阻尼比” $\xi=0.707$ (ω_n 确定时, t_s 实际最小)

- 极点实部 $\xi\omega_n=C$ 时

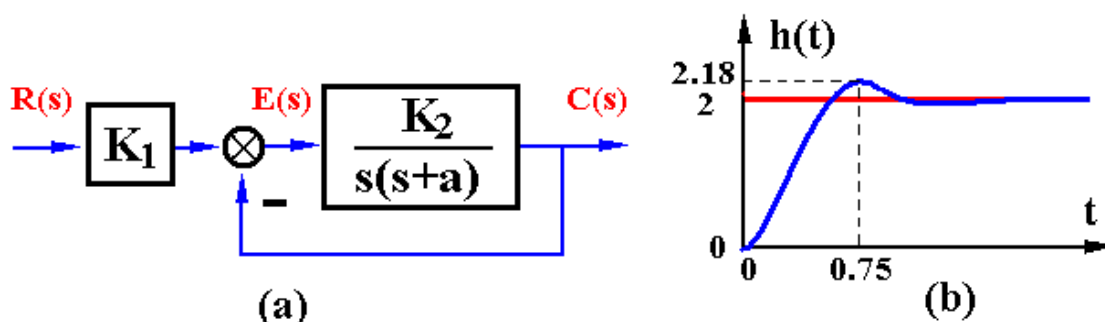


- 无阻尼自然频率 $\omega_n=C$ 时



例 2 二阶系统的结构图及单位阶跃响应分别如图(a), (b)所示。

试确定系统参数 K_1, K_2, a 的值。



解 由结构图可得系统闭环传递函数

$$\Phi(s) = \frac{K_1 K_2 / s(s+a)}{1 + K_2 / s(s+a)} = \frac{K_1 K_2}{s^2 + as + K_2} \quad \begin{cases} K_2 = \omega_n^2 \\ a = 2\xi\omega_n \end{cases}$$

由单位阶跃响应曲线有

$$h(\infty) = 2 = \lim_{s \rightarrow 0} s\Phi(s)R(s) = \lim_{s \rightarrow 0} \frac{K_1 K_2}{s^2 + as + K_2} = K_1$$

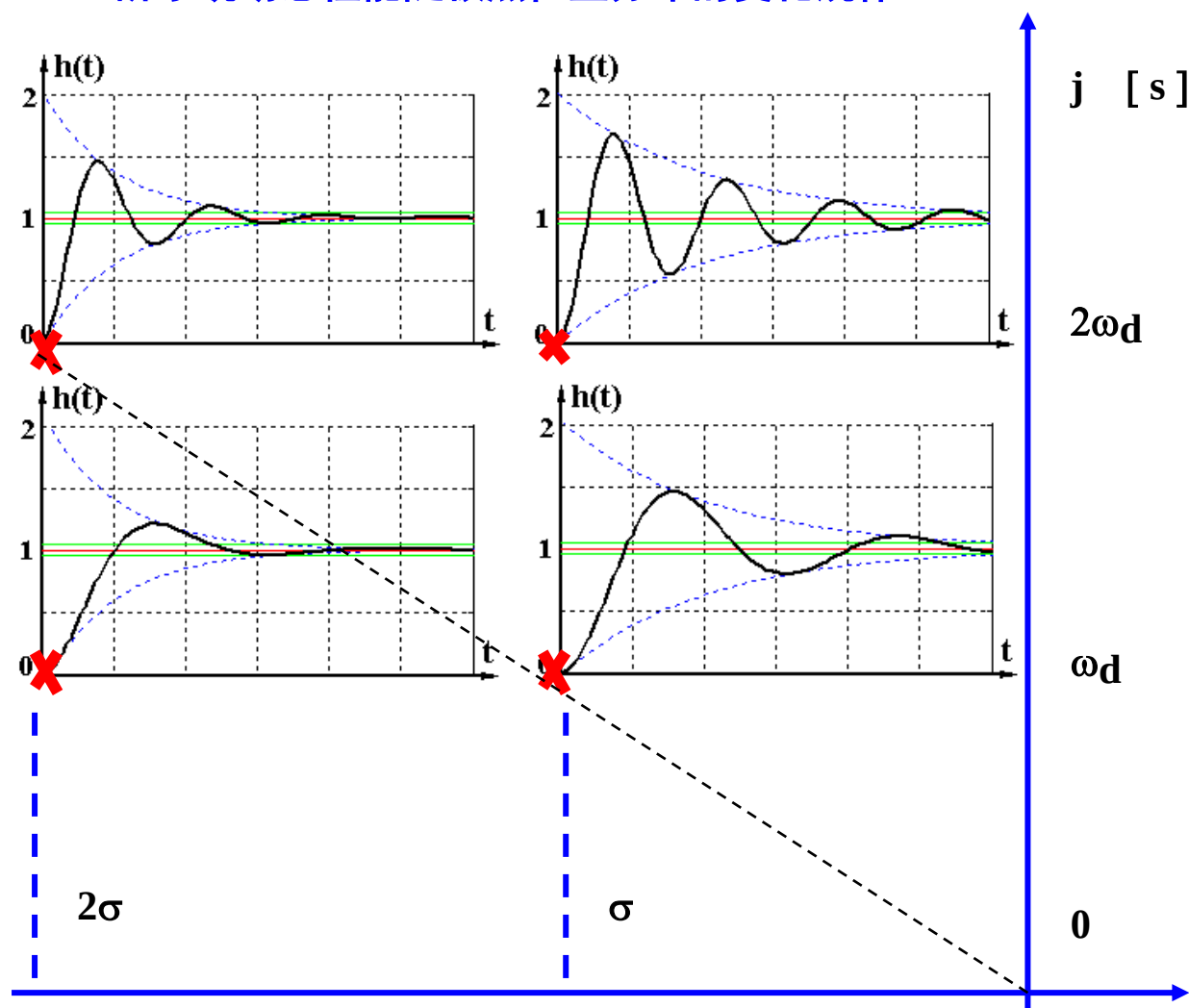
$$\begin{cases} t_p = \frac{\pi}{\sqrt{1-\xi^2}\omega_n} = 0.75 \\ \sigma\% = \frac{2.18-2}{2} = 0.09 = e^{-\xi\pi/\sqrt{1-\xi^2}} \end{cases}$$

联立求解得
$$\begin{cases} \xi = 0.608 \\ \omega_n = 5.278 \end{cases}$$

$$\begin{cases} K_2 = \omega_n^2 = 5.278^2 = 27.85 \\ a = 2\xi\omega_n = 2 \times 0.608 \times 5.278 = 6.42 \end{cases}$$

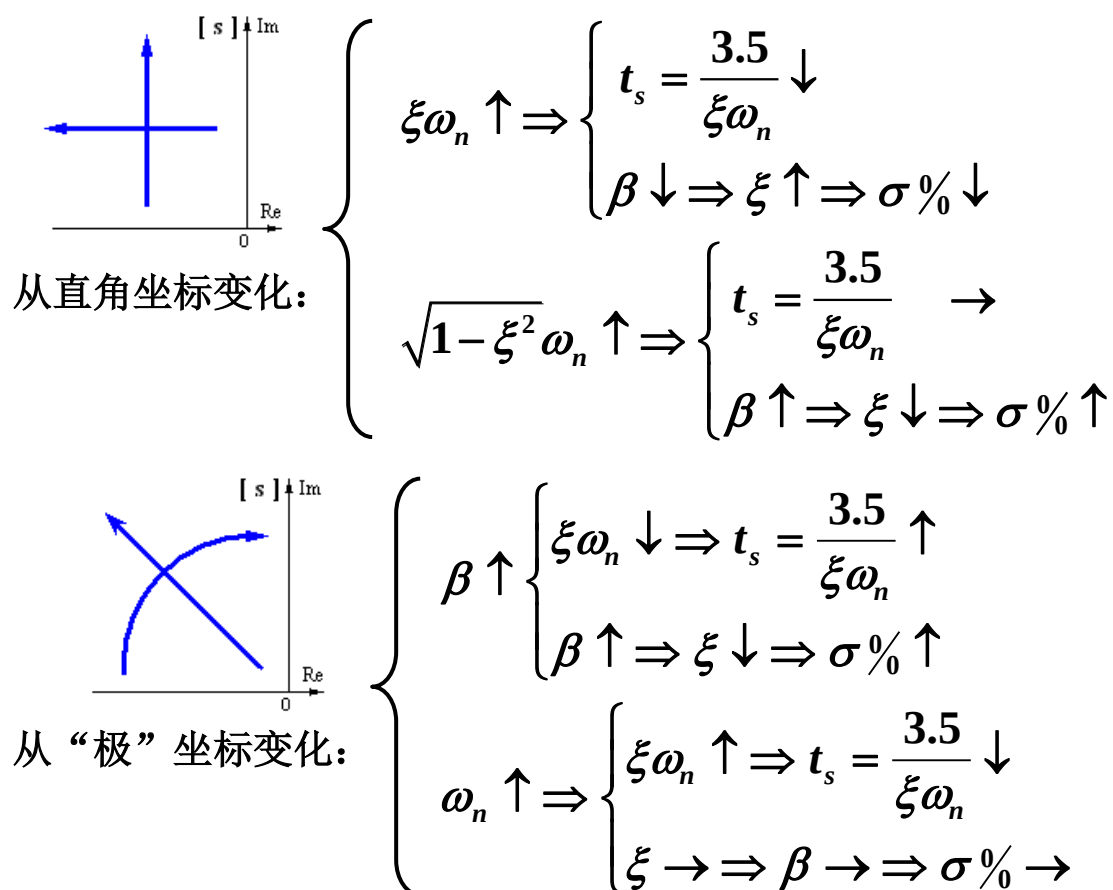
因此有 $K_1 = 2, K_2 = 27.85, a = 6.42$ 。

5. 二阶系统动态性能随极点位置分布的变化规律



[计算演示]

欠阻尼二阶系统动态性能随极点位置的变化规律小结



例 3 典型欠阻尼二阶系统

$$\text{要求 } \left\{ \begin{array}{l} 5\% \leq \sigma \% < 16.3\% \\ 2 < \omega_n \leq 5 \end{array} \right.$$

试确定满足要求的系统极点分布范围。

解. 依题意有

$$\left\{ \begin{array}{l} 0.707 \geq \xi > 0.5 \\ 2 < \omega_n \leq 5 \end{array} \right. \quad \left\{ \begin{array}{l} 45^\circ \leq \beta < 60^\circ \\ 2 < \omega_n \leq 5 \end{array} \right.$$

